Mass Scaling of Harmonic Oscillator Matrix Elements in Nuclear Physics

Morten Hjorth-Jensen

Notes from March 1996

Motivation

In shell-model calculations, effective two-body matrix elements are often constructed for a specific mass region.

Problem

Given matrix elements fitted or computed for a nucleus with mass number $A_0 = 100$, how can they be approximately scaled to a nucleus with $A = 132$?

The answer is connected to:

- harmonic oscillator basis functions,
- oscillator frequency $\hbar\omega$,
- oscillator length parameter b,
- phenomenological shell-model scaling laws.

Harmonic Oscillator Basis

Single-particle shell-model states are commonly expanded in a harmonic oscillator basis. The harmonic oscillator Hamiltonian is

$$H = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 r^2.$$

The corresponding oscillator length is

$$b = \sqrt{\frac{\hbar}{m\omega}}.$$

The radial wave functions therefore depend explicitly on the oscillator frequency. Changing the mass number A changes the optimal value of $\hbar\omega$ and thus the spatial extent of the basis.

Empirical Oscillator Frequency

A widely used empirical formula is

$$\hbar\omega \approx 41A^{-1/3}\ \mathrm{MeV}.$$

A more refined formula is

$$\hbar\omega \approx 45A^{-1/3} - 25A^{-2/3}\ \mathrm{MeV}.$$

For the nuclei considered here:

$$\hbar\omega(100) \approx 8.53\ \mathrm{MeV}, \qquad \hbar\omega(132) \approx 7.87\ \mathrm{MeV}.$$

Hence

$$\frac{\hbar\omega(132)}{\hbar\omega(100)} \approx 0.923.$$

Oscillator Length Scaling

Since

$$b = \sqrt{\frac{\hbar}{m\omega}},$$

we obtain

$$b \propto \omega^{-1/2}.$$

Using the empirical scaling $\omega \propto A^{-1/3}$, we find

$$b \propto A^{1/6}.$$

Therefore

$$\frac{b_{132}}{b_{100}} = \left(\frac{132}{100}\right)^{1/6} \approx 1.047.$$

Interpretation of Oscillator-Length Scaling

The scaling

$$\frac{b_{132}}{b_{100}} \approx 1.047$$

means that the oscillator basis for $A = 132$ is about 4.7% more spatially extended than that for $A = 100$.

Interpretation:

- The basis for $A = 132$ is spatially more extended.
- Wave functions become slightly broader.
- Overlap integrals are modified.
- Two-body matrix elements change through the basis parameter.

Phenomenological Scaling of Matrix Elements

A commonly used phenomenological scaling law for shell-model two-body matrix elements is

$$V_{JT}(A) = V_{JT}(A_0)\left(\frac{A}{A_0}\right)^{-\alpha}, \qquad \alpha \approx 0.3.$$

For the present example:

$$V(A=132) = V(A=100)\left(\frac{132}{100}\right)^{-0.3}.$$

This gives

$$V(A=132) \approx 0.92\, V(A=100).$$

Physical Interpretation

Why does the interaction strength decrease slightly?

As the nucleus becomes larger:

- the oscillator length b increases,
- single-particle wave functions spread out,
- overlap integrals decrease slightly,
- effective interaction matrix elements become weaker.

The scaling approximately captures changes in nuclear radius, basis dependence, renormalization effects, and effective many-body correlations.

More Fundamental Treatment

The phenomenological scaling law is only approximate.

A more rigorous approach would:

1. recompute the harmonic oscillator basis with the new b value,
2. recalculate Talmi integrals,
3. reevaluate the two-body matrix elements.

This is especially important for finite-range interactions, chiral EFT interactions, large changes in mass number, and precision spectroscopy.

Talmi–Moshinsky Formalism

Two-body matrix elements in the oscillator basis are often expressed as

$$\langle ab|V|cd\rangle = \sum_N C_N I_N.$$

Here

- C_N are Moshinsky brackets,
- I_N are Talmi integrals.

The Talmi integrals depend explicitly on the oscillator length:

$$I_N = I_N(b).$$

Therefore changing A changes the matrix elements through the basis parameter b.

Example: Scaling from $A = 100$ to $A = 132$

Using the phenomenological prescription

$$V(A) = V(A_0) \left(\frac{A}{A_0} \right)^{-0.3},$$

we obtain

$$V(132) = V(100) \left(\frac{132}{100} \right)^{-0.3}.$$

Numerically,

$$\left(\frac{132}{100} \right)^{-0.3} \approx 0.92.$$

Hence

$$V(132) \approx 0.92\, V(100),$$

which corresponds to a reduction of approximately 8%.

Typical Usage in Shell-Model Calculations

Mass scaling is commonly employed in:

- USD interactions,
- large-scale shell-model calculations,
- effective interactions derived from G-matrix theory.

Advantages:

- simple implementation,
- captures average trends,
- computationally inexpensive.

Limitations of Simple Mass Scaling

The same scaling has important limitations:

- it is not fully microscopic,
- it cannot replace full renormalization,
- it may fail far from fitted mass regions,
- it cannot capture detailed monopole and multipole retuning.

For high-precision spectroscopy, the preferred procedure is to recompute or refit the effective interaction in the relevant mass region and oscillator basis.

Summary: Scaling Relations

The central scaling relations are:

$$\hbar\omega \propto A^{-1/3}, \qquad b \propto A^{1/6}.$$

Effective shell-model matrix elements are often scaled phenomenologically as

$$V(A) = V(A_0)\left(\frac{A}{A_0}\right)^{-0.3}.$$

For scaling from $A = 100$ to $A = 132$, this gives

$$V(132) \approx 0.92\, V(100).$$

Summary: Practical Recommendation

A simple first estimate is therefore to multiply the $A = 100$ two-body matrix elements by approximately 0.92.

However, a more accurate treatment requires recomputing the matrix elements using the new oscillator basis. Reference [2] shows how this can be done using many-body theories.

References

I. Talmi,
Simple Models of Complex Nuclei, Harwood Academic Publishers (1993).

M. Hjorth-Jensen, T. T. S. Kuo, and E. Osnes,
Realistic effective interactions for nuclear systems, Physics Reports **261**, 125 (1995).